HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2025)

Student No.: 01 Name: _____ Marks: / 128

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. If $x = 5.67$ (correct to 2 decimal places), find the range of values of x.

   A. $5.66 < x \leq 5.68$

   B. $5.66 \leq x < 5.68$

   C. $5.665 < x \leq 5.675$

   D. $5.665 \leq x < 5.675$

(DSE-2025 PII Q06)

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2. If α, β and γ are non-zero constants such that $(\alpha+2\beta):(\beta+2\gamma):(\gamma+2\alpha)=4:9:5$, then $\alpha:\beta=$

A. $2:5$.

B. $5:2$.

C. $128:149$.

D. $149:128$.

(DSE-2025 PII Q11)

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3. Let a_n be the nth term of a sequence. If $a_2 = 3$, $a_5 = 41$ and $a_{n+2} = 2a_{n+1} + a_n$ for any positive integer n, then $a_6 =$

A. 99.

B. 101.

C. 239.

D. 243.

(DSE-2025 PII Q13)

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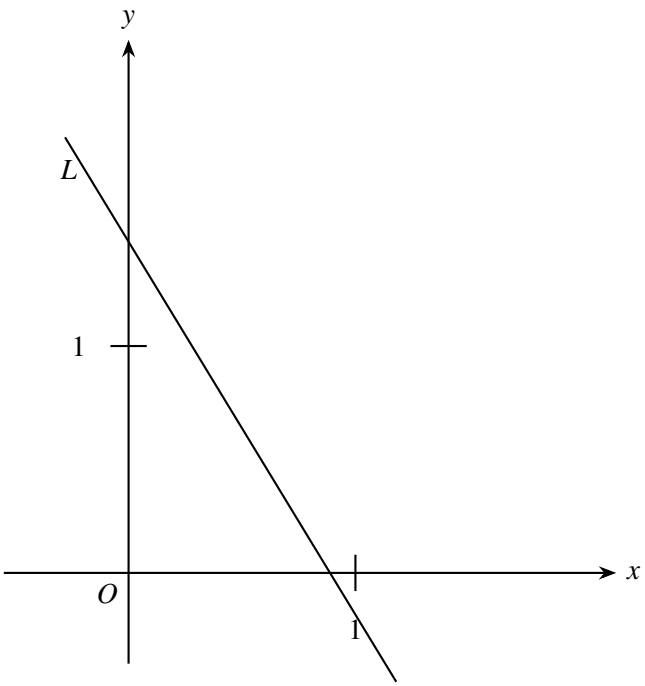
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4. In the figure, the equation of the straight line L is $px + qy = 7$. Which of the following is/are true?

I. $p > 7$

II. $q > 7$

III. $q > p$



A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2025 PII Q14)

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5. The perimeter of the sector OMN is 12π cm, where O is the centre of the sector OMN. If the radius of the sector OMN is 3π cm, which of the following are true?

I. The area of the sector OMN is $9\pi^2$ cm^2.

II. The perimeter of $\triangle OMN$ is less than 35 cm.

III. The angle of the sector OMN is greater than $100°$.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

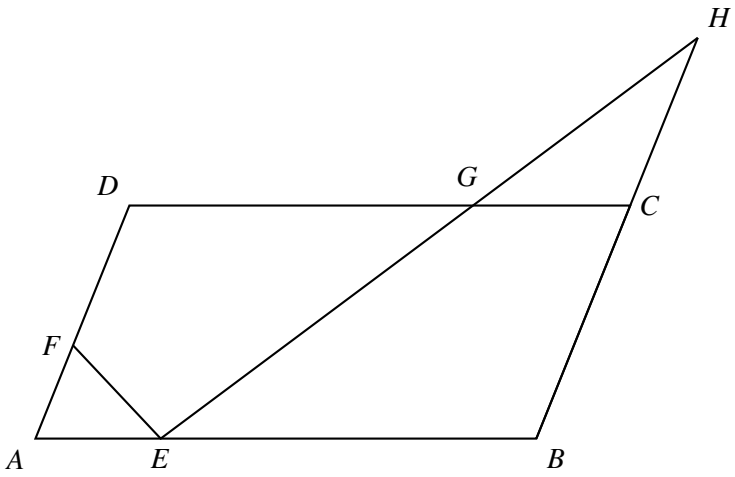
(DSE-2025 PII Q15)

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6. In the figure, $ABCD$ is a parallelogram. Let E, F and G be points lying on AB, AD and CD respectively such that $BE = 3AE$, $2DF = 3AF$ and $DG = 2CG$. It is given that BC produced and EG produced meet at the point H. If the area of $\triangle CGH$ is $16\,\text{cm}^2$, then the area of the quadrilateral $DFEG$ is



A. $46\,\text{cm}^2$.

B. $49\,\text{cm}^2$.

C. $105\,\text{cm}^2$.

D. $115\,\text{cm}^2$.

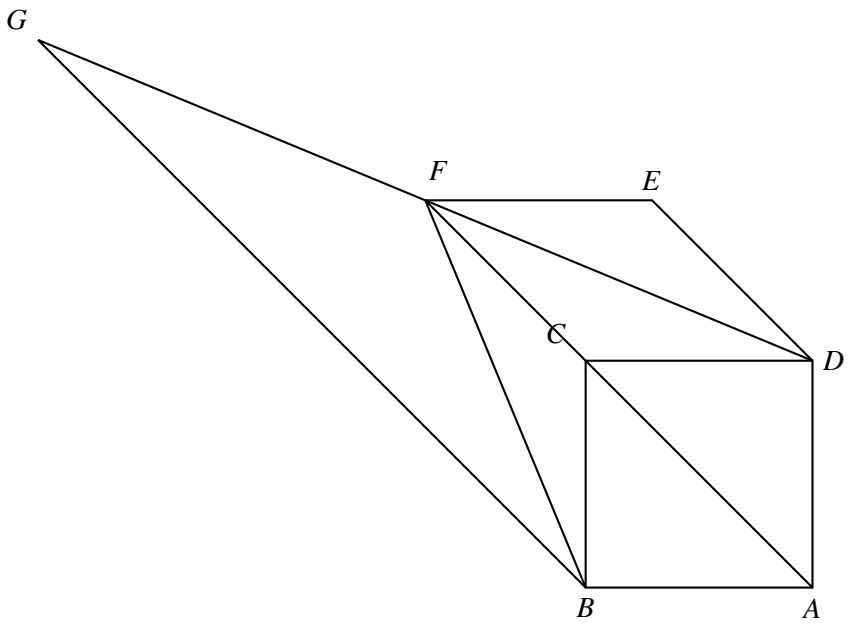
(DSE-2025 PII Q17)

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7. In the figure, $ABCD$ is a square. Let E be a point such that $CDEF$ is a rhombus, where F is a point lying on AC produced. DF is produced to the point G such that $AF \parallel BG$. Which of the following are true?



I. $DF = FG$

II. $\triangle BFG \sim \triangle DEF$

III. $\angle ABG + \angle BFD = 180^\circ$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2025 PII Q19)

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8. It is given that $PQRS$ is a trapezium with $PQ \,//\, SR$. If $PS = 41\,\text{cm}$, $RS = 53\,\text{cm}$, $\angle PSR = 120^\circ$ and $\angle QRS = 150^\circ$, then $PQ =$

A. 82 cm.

B. 100 cm.

C. 106 cm.

D. 135 cm.

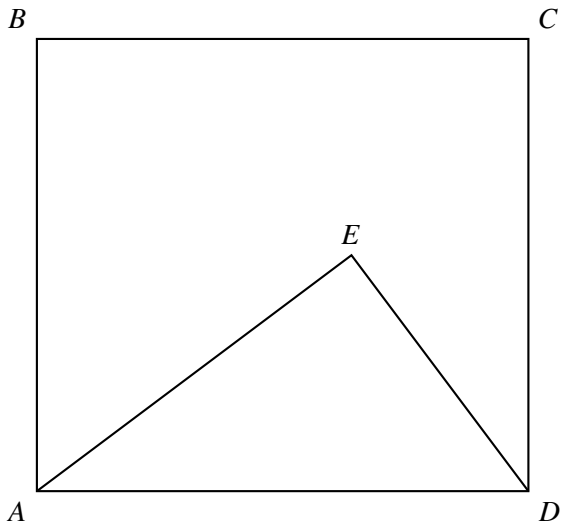
(DSE-2025 PII Q20)

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9. In the figure, $ABCD$ is a rectangle. It is given that $AE = 20\,\text{cm}$ and $\angle AED = 90°$. If the area of $\triangle ADE$ is $150\,\text{cm}^2$, then the perpendicular distance from E to CD is



A. 8 cm.

B. 9 cm.

C. 12 cm.

D. 15 cm.

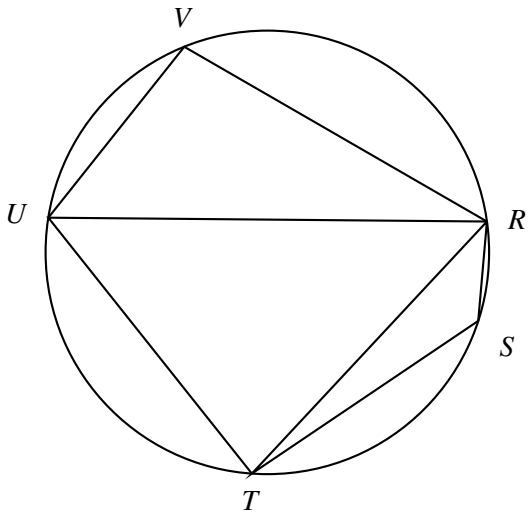
(DSE-2025 PII Q21)

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10. In the figure, $RSTUV$ is a circle. It is given that $RT \,//\, VU$ and RT is the angle bisector of $\angle SRU$. If $\angle RTS = \angle URV = 33°$, then $\angle RUT =$



A. 36°.

B. 49°.

C. 65°.

D. 71°.

(DSE-2025 PII Q22)

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11. Denote the origin by O. The coordinates of the point A are $(a, 2a)$, where $a > 0$. Let P be a moving point in the rectangular coordinate plane such that $AP = OA$. The locus of P is a

A. point.

B. circle.

C. rhombus.

D. straight line.

(DSE-2025 PII Q25)

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12. The straight lines L_1 and L_2 are perpendicular to each other. The equations of L_1 and L_2 are $3x+4y-20=0$ and $mx+ny-20=0$ respectively, where m and n are constants. Suppose that L_1 and L_2 cut the x-axis at the points A and B respectively. Let C be the point of intersection of L_1 and L_2. It is given that C lies above the x-axis. If the area of $\triangle ABC$ is 6, find n.

A. -12

B. -9

C. 9

D. 12

(DSE-2025 PII Q26)

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13. The coordinates of the centre of the circle C are $(7,-5)$. If C cuts the x-axis at the points P and Q such that $PQ = 24$, then the equation of C is

A. $x^2+y^2-14x+10y-95=0$.

B. $x^2+y^2+14x-10y-95=0$.

C. $x^2+y^2-14x+10y-119=0$.

D. $x^2+y^2+14x-10y-119=0$.

(DSE-2025 PII Q27)

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14. $3E000000000000_{16} =$

A. $2^{16}+2^{15}+2^{14}+2^{13}+2^{12}$.

B. $2^{17}+2^{16}+2^{15}+2^{14}+2^{13}$.

C. $2^{52}+2^{51}+2^{50}+2^{49}+2^{48}$.

D. $2^{53}+2^{52}+2^{51}+2^{50}+2^{49}$.

(DSE-2025 PII Q31)

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15. On the same rectangular coordinate system, the graph of $y = \log_a x$ cuts the graph of $y = a^x$ and the x-axis at the points P and Q respectively, where a is a positive constant. Denote the origin by O. Which of the following are true?

I. $a < 1$

II. $OQ > a$

III. $\angle POQ = 45°$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2025 PII Q34)

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16. $i^9 + i^{10} + i^{11} + \cdots + i^{999} =$

A. -1.

B. 0.

C. 1.

D. i.

(DSE-2025 PII Q35)

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17. Consider the following system of inequalities:

$$\begin{cases} x \leq 11 \\ 4x + 5y - 19 \geq 0 \\ 7x - 6y + 11 \leq 0 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. If (x, y) is a point lying in D, then the greatest value of $8x - 6y + 11$ is

A. 1.

B. 11.

C. 15.

D. 129.

(DSE-2025 PII Q36)

Answers written in the margins will not be marked.

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18. Let p, q and r be non-zero real numbers. If p, q, r is an arithmetic sequence, which of the following must be true?

I. 3^p, 3^q, 3^r is a geometric sequence.

II. $\frac{5}{p}$, $\frac{5}{q}$, $\frac{5}{r}$ is a geometric sequence.

III. $p-q$, $q-r$, $r-p$ is an arithmetic sequence.

A. I only

B. II only

C. I and III only

D. II and III only

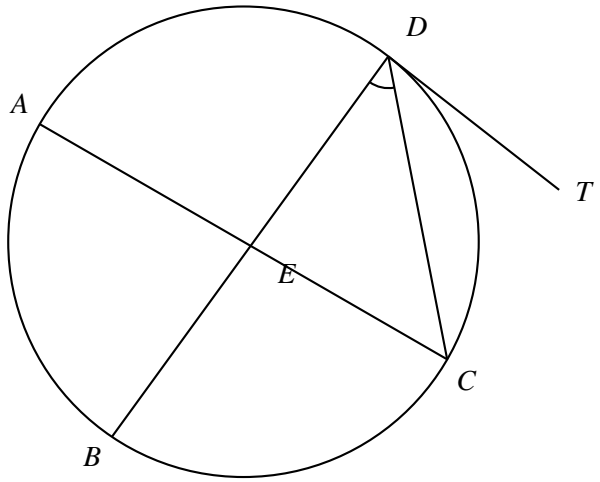
(DSE-2025 PII Q37)

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19. In the figure, AC is a diameter of the circle $ABCD$. Denote the point of intersection of AC and BD by E. It is given that TD is the tangent to the circle at D. If $\angle BEC = 96°$ and $\angle CDT = 41°$, then $\angle CDE =$



A. 47°.

B. 48°.

C. 52°.

D. 55°.

(DSE-2025 PII Q38)

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20. For $90° < \theta < 270°$, how many roots does the equation $\tan^3\theta = 2\tan\theta$ have?

A. 2

B. 3

C. 4

D. 5

(DSE-2025 PII Q39)

Answers written in the margins will not be marked.

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21. $PQRS$ is a regular tetrahedron. Find the angle between PQ and $\triangle QRS$ correct to the nearest degree.

A. 35°

B. 55°

C. 60°

D. 71°

(DSE-2025 PII Q40)

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22. There are 9 cans of apple juice and 4 cans of grape juice in a bag. If 6 cans are randomly chosen from the bag at the same time, find the probability that at most 3 cans of grape juice are chosen.

A. $\frac{9}{13}$

B. $\frac{133}{143}$

C. $\frac{140}{143}$

D. $\frac{714}{715}$

(DSE-2025 PII Q43)

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23. The standard scores of a boy and a girl in a Mathematics test are -2 and z respectively. The standard deviation of the scores of the Mathematics test is 2 marks. If the difference of the test score of the boy and the test score of the girl is 6 marks, find z.

A. -5 or 1

B. -5 or 3

C. -3 or 1

D. -3 or 3

(DSE-2025 PII Q44)

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24. It is given that a, b, c and d are four distinct real numbers. Let m_1, r_1 and v_1 be the mean, the range and the variance of the group of numbers $\{a,b,c,d\}$ respectively while m_2, r_2 and v_2 be the mean, the range and the variance of the group of numbers $\{2a,2b,2c,2d\}$ respectively. Denote the mean, the range and the variance of the group of numbers $\{a+3,b+3,c+3,d+3\}$ by m_3, r_3 and v_3 respectively. Which of the following are true?

I. $m_1+m_3>m_2$

II. $r_1+r_3=r_2$

III. $v_1+v_3<v_2$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2025 PII Q45)

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25. Simplify $\frac{(x^4y^{-5})^3}{xy^2}$ and express your answer with positive indices. (3 marks) (DSE-2025 PI Q01)

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26. The coordinates of the point A are $(4, -2)$. A is rotated clockwise about the origin O through 90° to B.

(a) Write down the coordinates of B.

(b) Suppose that B is translated upwards by t units to the point C. If A, O and C are collinear, find t.

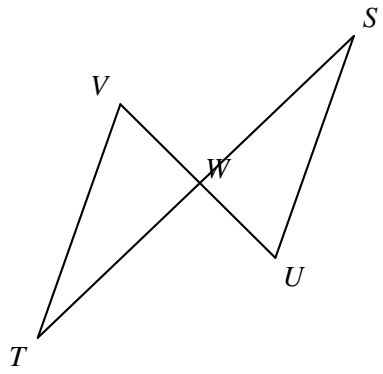
(3 marks) (DSE-2025 PI Q04)

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27. In the figure, ST and UV intersect at the point W. It is given that $SU \parallel VT$ and W is the mid-point of ST.



(a) Prove that $\triangle SUW \cong \triangle TVW$.

(b) Let X be a point lying on TW such that $\triangle SUW \sim \triangle VXW$. If $SU = 57\,\text{cm}$, $SW = 63\,\text{cm}$ and $WX = 7\,\text{cm}$, find the perimeter of $\triangle TVX$.

(5 marks) (DSE-2025 PI Q08)

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28. The table below shows the distribution of the numbers of pens owned by some students.

Number of pens	5	6	7	8	9
Number of students	9	13	2	s	11

The median of the distribution is 7.

(a) Find the least possible value of s.

(b) Write down the greatest possible value of s.

(c) Find the greatest possible standard deviation of the distribution.

(5 marks) (DSE-2025 PI Q09)

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29. Let $f(x) = 2x^3 + hx^2 + kx + 15$, where h and k are constants. When $f(x)$ is divided by $x+2$, the remainder is -45. It is given that $f(x)$ is divisible by $2x-5$.

(a) Find h and k.

(b) Someone claims that all the roots of the equation $f(x) = 0$ are rational numbers. Do you agree? Explain your answer.

(6 marks) (DSE-2025 PI Q10)

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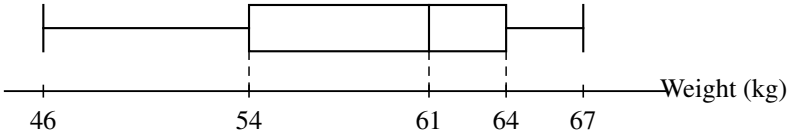
30. The stem-and-leaf diagram below shows the distribution of the weights (in kg) of some athletes before training.

Stem (tens)	Leaf (units)
4	5 6 8
5	w w 6 7
6	3 4 5 6 9 9
7	0 2
8	7

The difference of the range and the inter-quartile range of the above distribution is 25 kg.

(a) Write down the range of the above distribution. Hence, find w.

(b) The box-and-whisker diagram below shows the distribution of the weights (in kg) of the athletes after the training.



(i) Find the change in the upper quartile of the distribution due to the training.

(ii) Is the distribution of the weights of the athletes after the training less dispersed than that before the training? Explain your answer.

(7 marks) (DSE-2025 PI Q12)

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31. The straight line L cuts the positive x-axis and the positive y-axis at the points M and N respectively. Let P be a moving point in the rectangular coordinate plane such that $MP = NP$. Denote the locus of P by Γ. It is given that the equation of Γ is $3x - 2y - 30 = 0$.

(a) Describe the geometric relationship between Γ and MN.

(b) Find the equation of L.

(c) Suppose that Γ cuts L and the x-axis at the points Q and R respectively. Find the area of the quadrilateral $ONQR$, where O is the origin.

(7 marks) (DSE-2025 PI Q13)

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32. The height and the base radius of a solid metal right circular cone are 45 cm and 24 cm respectively. The circular cone is divided into a circular cone and a frustum X by a plane parallel to its base. It is given that the height of X is 30 cm.

(a) Express the volume of X in terms of π.

(b) Express the total surface area of X in terms of π.

(c) X is melted and recast into a solid cube. Does the total surface area of the cube exceed the total surface area of X? Explain your answer.

(9 marks) (DSE-2025 PI Q14)

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33. There are 8 white plates, 4 red plates and 1 black plate in a box. If 3 plates are randomly drawn from the box at the same time, find

(a) the probability that the colours of the 3 plates drawn are the same;

(b) the probability that the colours of the 3 plates drawn are all different.

(4 marks) (DSE-2025 PI Q15)

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34. It is given that $\begin{cases} \log_3 x + \log_3 y = 9 \\ \log_x 81 - \log_y 9 = 1 \end{cases}$, where $0 < x < y$.

(a) Let $u = \log_3 y$. Prove that $u^2 - 3u - 18 = 0$.

(b) Find x.

(4 marks) (DSE-2025 PI Q16)

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35. Let $T(n)$ be the nth term of an arithmetic sequence such that $T(1) \neq T(2)$ and $T(47) = 456$. It is given that $T(9)$, $T(47)$, $T(199)$ is a geometric sequence.

(a) Find $T(1)$.

(b) Find the least value of n such that the sum of the first n terms of the arithmetic sequence is greater than 10^6.

(6 marks) (DSE-2025 PI Q17)

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36. Let $g(x) = 3x^2 - 6kx + 24x + 3k^2 - 24k + 55$, where k is a positive constant. Denote the vertex of the graph of $y = g(x)$ by R.

(a) Using the method of completing the square, express the coordinates of R in terms of k.

(b) The graph of $y = h(x)$ is obtained by translating the graph of $y = g(x)$ rightwards by 6 units and then downwards by 10 units. Denote the vertex of the graph of $y = h(x)$ by S. Let T be a point such that the orthocentre of $\triangle RST$ is the origin.

(i) Express the coordinates of T in terms of k.

(ii) The coordinates of the point U are $(-5, 5)$. Let V be a point lying on RS such that TV is perpendicular to RS. It is given that S, T, U and V are concyclic. Find k.

(9 marks) (DSE-2025 PI Q18)

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37. The equation of the straight line L is $4x-3y+83=0$. The equation of the circle C is $x^2+y^2+ax-2y+b=0$, where a and b are positive constants. It is given that C passes through the point $(-10,9)$ and L is a tangent to C.

(a) Prove that the equation $25x^2+(9a+640)x+(90a+4924)=0$ has equal roots. Hence, find a and b.

(b) The coordinates of the point P are $(-20,1)$. Let Q be a point lying below L such that PQ is tangent to C and $PQ=25$. Denote the centre of C by I.

(i) Find $\cos\angle IPQ$.

(ii) Find, in surd form, the distance between I and Q.

(iii) Let R be a point lying on L such that C is the inscribed circle of $\triangle PQR$. Denote the radius of the circumcircle of $\triangle PQR$ by r. Someone claims that $r>PQ$. Is the claim correct? Explain your answer.

(12 marks) (DSE-2025 PI Q19)

I can finish _____ questions

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